Lecture 2: Analytical Methods

Introduction to Data Analysis Techniques

1 Data Representation

Definition 1.1 (Finite Index Set Generator)**.** [def:finite-index-set] The finite index set generator is the function

$$\mathrm{Fin} : \mathbb{N} \to \mathcal{P}(\mathbb{N}) \equiv n \mapsto \{i \in \mathbb{N} \mid i < n\}$$

Thus $\mathrm{Fin}(n)$ contains the first n natural numbers, starting at 0; in particular, $\mathrm{Fin}(0) = \emptyset$ and $\mathrm{Fin}(3) = \{0, 1, 2\}$

Definition 1.2 (Pairing Function Generator)**.** [def:pairing-function-generator] The pairing function generator takes $n \in \mathbb{N}$ and returns an operation combining two indexed families $x, y : \mathrm{Fin}(n+1) \to \mathbb{R}$ into one indexed family with values in $\mathbb{R} \times \mathbb{R}$, representing $n + 1$ observations

$$\begin{aligned}\mathrm{pairGen} : \Pi n : \mathbb{N}.\, (\mathrm{Fin}(n+1) \to \mathbb{R}) &\to (\mathrm{Fin}(n+1) \to \mathbb{R}) \\ &\to \mathrm{Fin}(n+1) \to (\mathbb{R} \times \mathbb{R}) \\ &\equiv n \mapsto \big(x \mapsto \big(y \mapsto \big(i \mapsto (x(i), y(i))\big)\big)\big)\end{aligned}$$

Definition 1.3 (Pairing of Functions)**.** [def:pairing-functions] For a fixed $n \in \mathbb{N}$, define the pairing function by applying the pairing function generator to n

$$\begin{aligned}\mathrm{pair}_n : (\mathrm{Fin}(n+1) \to \mathbb{R}) &\to (\mathrm{Fin}(n+1) \to \mathbb{R}) \\ &\to \big(\mathrm{Fin}(n+1) \to (\mathbb{R} \times \mathbb{R})\big) \\ &\equiv \mathrm{pairGen}(n)\end{aligned}$$

For data fitting, $x(i)$ represents the input coordinate of observation i, and $y(i)$ represents its observed output, so

$$\mathrm{pair}_n(x)(y)(i) = (x(i), y(i))$$

Definition 1.4 (Distinct Input Families)**.** [def:distinct-input-families] The function DistinctFamily assigns to each $n \in \mathbb{N}$ the set of indexed families $x : \mathrm{Fin}(n+1) \to \mathbb{R}$ with pairwise distinct values

$$\begin{aligned}&\mathrm{DistinctFamily} : \Pi n : \mathbb{N}.\, \mathcal{P}(\mathrm{Fin}(n+1) \to \mathbb{R}) \\ &\qquad \equiv n \mapsto \left\{ x : \mathrm{Fin}(n+1) \to \mathbb{R} \;\middle|\; \begin{array}{c} \forall i, k \in \mathrm{Fin}(n+1), \\ i \neq k \implies x(i) \neq x(k) \end{array} \right\}\end{aligned}$$

Thus $x \in \mathrm{DistinctFamily}(n)$ means that x is injective

Convention 1.1 (Finite Index Ranges)**.** [conv:finite-index-sum-product-notation] Real Analysis prerequisite: generalize finite summation and finite products to families defined only on a finite interval of natural numbers, allowing index 0 and empty intervals. Only indices in the interval should be evaluated; retain 0 for an empty sum and 1 for an empty product

2 Lagrange Interpolation

Definition 2.1 (Interpolation)**.** [def:interpolation] Methods to find a smooth function $f(x)$ that passes through a given set of points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$

Definition 2.2 (Lagrange Interpolation). [def:lagrange-interpolation] For $n \in \mathbb{N}$, take

$$x \in \text{DistinctFamily}(n), \qquad y : \text{Fin}(n+1) \to \mathbb{R}$$

Lagrange interpolation for the paired observations $\text{pair}_n(x)(y)$ seeks a polynomial $f : \mathbb{R} \to \mathbb{R}$ of degree at most n such that

$$\forall i \in \text{Fin}(n+1), \quad f(x(i)) = y(i)$$

The values $y(i)$ may repeat

Theorem 2.1 (Unique Existence of Lagrange Interpolation). [thm:lagrange-existence-uniqueness] *For any $n+1$ data points $(x_0, y_0), \ldots, (x_n, y_n)$ with pairwise distinct $x_0, \ldots, x_n$, there exists a unique polynomial of degree at most n that passes through all these points*

Convention 2.1 (Lagrange Interpolation Method). [conv:lagrange-interpolation-method] For data satisfying the Lagrange interpolation conditions, write $x_i \equiv x(i)$ and $y_i \equiv y(i)$ and construct the following system of equations:

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + \ldots + a_n x_0^n &= y_0 \\ a_0 + a_1 x_1 + a_2 x_1^2 + \ldots + a_n x_1^n &= y_1 \\ \vdots & \vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \ldots + a_n x_n^n &= y_n \end{cases} \implies \begin{bmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & \ldots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

According to the Theorem in Linear Algebra , which states that a system of n linear equations with n unknowns has a unique solution if $\det A \neq 0$, where A is the coefficient matrix.
Since $x_0, \ldots, x_n$ are pairwise distinct, $\det A \neq 0$. Therefore, there exists a unique solution $\vec{a}$
So, we can find a unique polynomial $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$ that passes through all the given points.

Definition 2.3 (Lagrange Basis Polynomial Generator). [def:lagrange-base-polynomials-generator]

$$\text{LagrangeTerm} : \Pi n : \mathbb{N}.\, \text{DistinctFamily}(n) \to \text{Fin}(n+1) \to (\mathbb{R} \to \mathbb{R})$$
$$\equiv n \mapsto x \mapsto j \mapsto \Bigg(t \mapsto \prod_{0}^{j-1} \left(i \mapsto \frac{t - x(i)}{x(j) - x(i)} \right) \cdot \prod_{j+1}^{n} \left(i \mapsto \frac{t - x(i)}{x(j) - x(i)} \right) \Bigg)$$

Here n is the largest data index, j selects the basis polynomial, and x supplies the input coordinates. The output is a function of $t \in \mathbb{R}$, using the finite sum and product notation for the two index ranges

Redirect (Lagrange Basis Notation) [def:lagrange-base-polynomials]
Moved to convention 2.2.

Convention 2.2 (Lagrange Basis Notation). [conv:lagrange-base-polynomials] For fixed $n \in \mathbb{N}$ and $x \in \text{DistinctFamily}(n)$, write

$$L \equiv \text{LagrangeTerm}(n)(x), \qquad L_j \equiv L(j) \quad (j \in \text{Fin}(n+1))$$

Definition 2.5 (Lagrange Polynomial Generator). [def:lagrange-polynomial-generator]

$$\text{LagrangePoly} : \Pi n : \mathbb{N}.\, \text{DistinctFamily}(n) \to (\text{Fin}(n+1) \to \mathbb{R}) \to (\mathbb{R} \to \mathbb{R}) \equiv$$
$$n \mapsto x \mapsto y \mapsto t \mapsto \sum_{0}^{n} (j \mapsto y(j) \cdot \text{LagrangeTerm}(n)(x)(j)(t))$$

Definition 2.6 (Lagrange Interpolating Polynomial)**.** [def:lagrange-interpolating-polynomial] For $n \in \mathbb{N}$, $x \in \text{DistinctFamily}(n)$, and $y : \text{Fin}(n+1) \to \mathbb{R}$, the Lagrange interpolating polynomial is

$$f : \mathbb{R} \to \mathbb{R} \equiv t \mapsto \text{LagrangePoly}(n)(x)(y)(t)$$

Theorem 2.2 (Interpolation at the Data Points)**.** [thm:lagrange-interpolation-values] *Let $n \in \mathbb{N}$, $x \in \text{DistinctFamily}(n)$, and $y : \text{Fin}(n+1) \to \mathbb{R}$, and let f be the Lagrange interpolating polynomial for these inputs. Then*

$$\forall i \in \text{Fin}(n+1), \quad f(x(i)) = y(i)$$

Proof. **TODO** □

3 Linear Least Squares Fit

4 General Least Square Fit